VUG (6)—BCA (C-6001)

2020-23

Full Marks : 70

Time : 3 hours

Answer from **both** the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions : 10 × 4

1. Define Operation Research. Also explain the main characteristics and scope of OR.

2. Define Linear Programming Problem. Also discuss the mathematical structure of LPP.

(*Turn Over*)

3. Use the graphical method to solve the following LP Problem :

Maximize $Z = 2x_1 + x_2$

Subject to constraints

$x_1 + 2x_2 \le 10$ — (I)

$x_1 + x_2 \le 6$ — (II)

$x_1 - x_2 \le 2$ — (III)

$x_1 - 2x_2 \le 1$ — (IV)

and $x_1, x_2 \ge 0$

4. Use simplex method to solve the following LP Problem :

Max $Z = 4x_1 + 3x_2$

S to C

$2x_1 + x_2 \le 1000$

$x_1 + x_2 \le 800$

$x_1 \le 400$

$x_2 \le 700$

and $x_1, x_2 \ge 0$

$x_1 = 0,\ x_2 = -0.5$

optimum

optimium

Simula

5. Explain Big-M method and its procedure to solve LPP.

6. Determine an initial basic feasible solution to the following transportation problem by using NWCM and LCM method.

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	

7. A computer centre has three expert programmes. The centre wants three application programmers to be developed. The head of the computer centre after studying carefully the programmes to be developed, estimates the computer time in minutes required by the experts for the application programmes as follows :

		Programmer		
		A	B	C
Programmes	1	120	100	80
	2	80	90	110
	3	110	140	120

Assign the programmers to the programmes in such a way that the total computer time is minimum.

8. Discuss different phases of project management. Also explain procedure to design Network diagram.

GROUP—B

All questions are compulsory : 3 × 10

9. Explain feasible region.

10. Define objective function.

11. Discuss non-negativity condition in LPP.

12. Discuss the following :
    (*a*) Artificial variable
    (*b*) Slack variable
    (*c*) Surplus variable

13. What are the application areas of linear programming ?

14. What do you mean by unbounded solution ?

15. Explain Transportation problem.

16. Discuss the basic difference between PERT and CPM.

17. How we design network diagram in PERT/CPM ?

18. Define Assignment Problem.

VUG(6)-BCA(C-6002)

2020-23

Full Marks : 70

Time : 3 hours

Answer from **all** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION–A

Answer any *four* questions: 10 × 4

1. What is management ? What are the objectives of management ? Discuss.

2. Write down about the functions of management.

3. Discuss about "ELTON MAYO THEORY".

4. What are the functions of management ? Explain each.

(Turn Over)

5. What are the types of organizations ? Discuss.

6. What are the barriers of effective communication ?

7. What are the types of motivation ? Explain each.

8. What is coordination ? What are the need for coordination ? Explain the characteristics of coordination.

SECTION–B

Answer *all* questions 3 × 10

9. Define management as art and science.

10. What is scientific management ?

11. Write about the management skills.

12. What is organization ?

13. What are the functions of leader ?

14. Write about the importance of staffing.

15. What is the meaning of Delegation ?

16. What is Ethics ?

17. What is planning ?

18. What do you mean by need of controlling?

2020-23

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Time : 3 hours

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SECTION–A

Answer any *four* questions: 10 × 4

1. Why accounting is needed ? State the advantages and disadvantages of accounting.

2. Explain the different users of accounting information.

3. What are the different classifications of account ? Explain each with suitable examples.

(*Turn Over*)

4. Explain the rules of debit and credit ? Also explain Double Entry System.

5. Distinguish between Journal and Ledger.

6. Explain the use of MIS in accounting.

7. Explain Fund Flow Statement and Cash Flow Statement.

8. Pass Journal Entries of Mr. Mohon for following transactions:

April	1	Introduce Capital in Cash	50,000/-
	5	Goods Purchase in Cash	2,000/-
	7	Goods purchase on Credit from Mr. Sohan	5,000/-
	9	Cash soles	1,000/-
	12	Purchased Computer	10,000/-
	15	Paid Rent	5,000/-
	17	Paid Salary	10,000/-
	23	Received Interest	2,000/-
	25	Sold goods for Cash	6,000/-
	29	Received Commission	500/-

SECTION—B

Answer *all* questions: 3 × 10

9. What are the benefits of subsidiary book?

10. What is Trade discount and Cash discount? Explain with example.

11. What is the use of Trial balance in accounting ?

12. What are the different types of Assets ?

13. What is the use of Tax accounting ?

14. What are stocks or Inventories ? Explain its types.

15. What do we prepare a balance sheet ? What are the different fields of a balance sheet?

16. What is the use of accounting standard ?

17. Explain the meaning and importance of financial management.

18. Explain the different functions of database in on accounting system.

2020-23

Full Marks : 70

Time : 3 hours

Answer from **all** the Groups as directed.

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GROUP—A

Answer any *four* questions: 10 × 4

1. Explain about the OSI security architecture.

2. Define various kinds of attacks.

3. What are the properties of a digital signature ?

4. Explain classical Encryption techniques.

(*Turn Over*)

5. What is firewall ? Explain the various types of firewall.

6. Explain the procedure for RSA cryto-system.

7. Write about electronic mail security.

8. Define Integrity and non-repudiation.

GROUP—B

Answer *all* questions: 3 × 10

9. Distinguish between active and passive attacks.

10. What is Kerbcros ?

11. Define cipher text with an example.

12. Explain about the Hash functions.

13. Define Encryption and Decryption.

14. Explain risk & vulnerability.

15. Define virus. Explain in detail.

16. Describe about trusted systems.

17. Explain private key and public key.

18. What is message authentication ?

VUG(6)-BCA (E-6008)

2020-23

Full Marks : 70

Time : 3 hours

Answer from **both** the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions 10 × 4

1. Write and explain forward and Backward reasoning.

2. Write about hill climbing techniques.

3. Explain semantic nets.

4. Write the algorithm of Best first search and explain.

(*Turn Over*)

5. Write 10 commands of LISP with example.

6. Write and explain structure of an expert system.

7. Write about the characteristics of A. I. Language.

8. Write and explain Depth first search with example.

GROUP—B

Answer *all* questions 3 × 10

9. What is heuristic search ?

10. What is conceptual dependency ?

11. What is predicate logic ?

12. Write about PROLOG.

13. What is knowledge base ?

14. What is expert system ?

15. What is agents ?

16. What is frames ?

17. What is A* search ?

18. Write about control strategies.